



Sparse Dictionary Learning for Interpretable Lung Disease Diagnosis from 3D Chest CT

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by

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Abstract

Abstract goes here.

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Acronyms

AI Artificial Intelligence

C/G Consolidation-to-Ground-Glass-Opacity

CNN Convolutional Neural Network

CT Computed Tomography

GGO Ground-Glass Opacity

HU Hounsfield Unit

KSVD K-Singular Value Decomposition

LC-KSVD Label-Consistent K-Singular Value Decomposition

Chapter 1

Introduction

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality. In the diagnosis of lung disease, assessing the pattern, location, and regional distribution of involvement is the domain of radiology. Clinical diagnosis increasingly relies on radiological patterns to differentiate between pathologies with overlapping symptoms. While Chest X-rays (CXR) remain the first-line screening tool due to low cost and radiation, they suffer from limited sensitivity (roughly 70% visibility of lung volume due to anatomical superposition, in which all anatomical structures located in the path of the X-rays are stacked). [1]

CT is a radiographic technology that combines X-ray equipment with a computer and a cathode ray tube display to produce images of cross sections of the human body. A Radiographic detector measures the X-ray profile. Inside the CT scanner, there is a rotating frame that has an X-ray tube mounted on one side and the detector mounted on the opposite side. A beam of X-ray is generated as a rotating frame spins the X-ray tube and detector around the patient as shown in figure (4). Each time the X-ray tube and detector make one complete rotation, an image or slice is acquired. As the X-ray tube and detector make this rotation, the detector takes numerous profiles of the attenuated X-ray beam. Each profile is reconstructed by the computer into a 2D image of the slice that was scanned. [2] The fundamental metric of a CT scan is the Hounsfield Unit (HU), a relative quantitative scale that describes radiodensity [1.1] In pulmonary imaging, HU values allow for the objective differentiation of tissues that might appear visually similar.

A key characteristic in CT is "ground-glass opacity" (GGO), which appears as hazy increased opacity of lung, with preservation of bronchial and vascular margins. It is caused by partial filling of airspaces, interstitial thickening (due to fluid, cells, and/or fibrosis), partial collapse of alveoli, increased capillary blood volume, or a combination of these. "Consolidation" on the other hand, appears as a homogeneous increase in pulmonary parenchymal attenuation that obscures the margins of vessels and airway walls (45) (Fig 19). An air bronchogram may be present. The attenuation characteristics of consolidated lung are only rarely helpful in differential diagnosis (eg, decreased attenuation in lipoid pneumonia [46] and increased in amiodaron toxicity. Consolidation is more opaque than GGO. While GGO often suggests a reversible or early-stage process (like early infection

or lepidic tumor growth), consolidation usually indicates a more aggressive or advanced stage (like bacterial pneumonia or invasive tumor components). The Consolidation-to-GGO (C/G) ratio is a quantitative metric calculated by dividing the volume or area of consolidation by the volume or area of ground-glass opacity within a lesion or throughout the lungs.

The traditional approach for CT analysis relies on a "slice-by-slice" approach where a radiologist views a series of axial cross-sections and identifies the "index slice"—the single image where a pathology appears at its largest diameter. A 2D slice only captures a small fraction of a 3D structure. If a nodule is irregularly shaped, the "largest" slice may not represent the true extent of the disease [2.2]. Studies show that 2D measurements are highly sensitive to the patient's orientation and the specific slice chosen, leading to high inter-observer variability [2.3]. 2D models treat each slice as an independent data point, losing the "spatial continuity" between adjacent tissues [1.4]. 3D volumetric analysis represents a quantitative shift wherein the entire dataset is used to reconstruct the lung's anatomy as a solid volume. ?? [5.1].

Modern radiology increasingly relies on artificial intelligence to streamline the acquisition of radiologist analyses on chest X-rays, as evidenced by a study wherein an AI system reduced interpretation delivery times from 11.2 days to a mere 2.7 days, reinforcing the potency of automated triaging systems in streamlining healthcare workflows and amplifying patient care standards [42].[1] However, the exponential increase in data volume generated by 3D CT scanners presents a profound challenge for human interpretation and computational analysis. While the prevailing research trend has focused heavily on the deployment of 3D deep learning architectures, such as Convolutional Neural Networks (CNNs), these models frequently encounter significant hurdles regarding computational overhead, data scarcity, and a critical lack of interpretability. The research proposed herein advocates for a departure from these "black box" paradigms, suggesting that sparse representation methods offer a superior framework for the volumetric identification of lung diseases by ensuring clinical interpretability.

1.2 Problem Statement

Modern radiology increasingly relies on artificial intelligence to manage the vast quantities of 3D data produced during routine screenings. 3D deep learning has been widely applied to the automatic diagnosis of lung diseases, achieving remarkable success in tasks such as segmenting suspected infection regions and differentiating between various types of carcinoma. Despite these advancements, the inherent complexity of 3D CNNs introduces a set of constraints that are often overlooked in the pursuit of high accuracy scores.

1.2.1 Computational Overhead

Processing a 3D CT scan involves treating stacked 2D slices as a single volumetric matrix, which demands significantly more GPU memory and processing power than standard image analysis. Unlike ordinary RGB images, CT scans consist of single-channel data with varying voxel dimensions and resolution, which com-

plicates spatial alignment and generalizability across different scanners. Furthermore, the "curse of dimensionality" remains a persistent threat; as the network depth increases to capture global features, the risk of gradient explosion and slow convergence becomes more pronounced.

1.2.2 Interpretability Bottleneck

Beyond computational limits, the clinical utility of deep learning is hampered by the interpretability paradox. While tools like Grad-CAM provide visual heatmaps to localize lesions, these post-hoc explanations are often insufficient for medical professionals who require a granular understanding of the model's decision-making process. In high-stakes environments, a model that shruggingly produces a diagnosis without an audit trail of its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance. This transparency gap has led to a renewed interest in "open book" models that are interpretable by design, where the internal logic is clear from the outset.

1.3 Objectives and Scope

1.3.1 Objectives

1. Develop a Dictionary Learning framework for Interpretable Lung Disease Classification from 3D Chest CT
 - Implement training in two phases for normal and pathological atoms
 - Integrate LC-KSVD with label consistency constraints
2. Visualize dictionary atoms and map them to anatomical features
 - Compare sparse coefficient patterns across disease classes
 - Demonstrate atom-to-pathology correspondence
3. Benchmark against baseline models
 - Compare with standard K-SVD without label consistency constraint
 - Compare with 3D CNN baseline (3D-ResNet)
4. Integrate clinical metrics for volumetric analysis
 - Implement HU-based segmentation for GGO and consolidation
 - Calculate C/G ratio and compare model predictions against clinical thresholds

1.3.2 Scope

Chapter 2

Literature Review

2.1 Previous Work

Discuss relevant previous work in the field. For instance, [?] discusses advancements in ML.

2.2 Gaps in Literature

Identify gaps in the existing literature that your project aims to address.

Chapter 3

Methodology

3.1 Research Design

Describe the research design and methodology. Use diagrams or flow charts to illustrate the methodology.

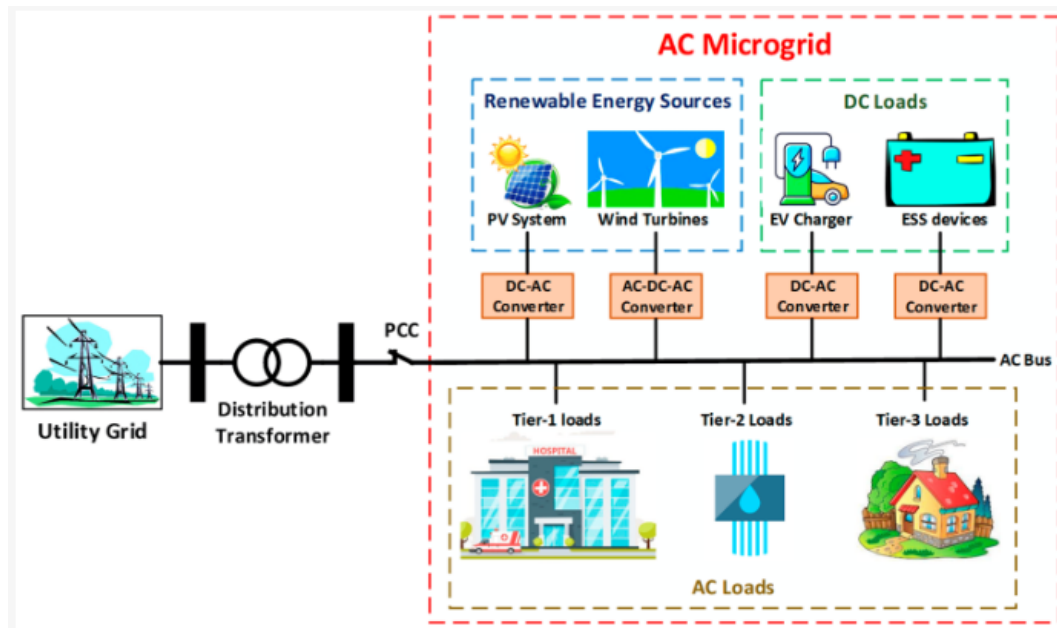


Figure 3.1: This is an example image.

As shown in Figure 3.1, the image is centered and has a caption.

3.2 Data Collection

Explain how data will be collected for the project.

Chapter 4

Timeline and Resource Required

4.1 Timeline

Provide a realistic timeline for completing the project. Use a Gantt chart or table.

4.2 Resource Required

State the resource required for the project and estimate the budget for the resources required.

Chapter 5

Conclusion

Summarize the key points of your proposal and reiterate the importance of the project.

References

- [1] A. Jones, *Machine Learning: Theory and Practice*. Tech Publications, 2019.